

Andrés Montoya Rojas

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EDUCATION

RWTH Aachen University | Oct 2019 – Nov 2025

Bachelor of Science in Computer Science

- Thesis: Portable and FAIR Query Functions with WebAssembly and SPARQL
- Interdisciplinary focus including psychology

Instituto Tecnológico de Costa Rica | Feb 2018 – Jun 2018

- Introduction to LaTeX and to programming in Python

TECHNICAL SKILLS

- Programming & Query Languages: Python, Java, JavaScript, SQL
- Libraries: scikit-learn, pandas
- Technologies & Tools: Git, WebAssembly, HTML, CSS

EXPERIENCE

Customer Churn Prediction Project | Python, scikit-learn, pandas

- Development of an end-to-end customer churn prediction pipeline including data auditing, cleaning, feature preprocessing, and binary classification on a real-world telecommunications dataset
- Evaluation and comparison of machine learning models (Logistic Regression and Random Forest) using train/test splits, preprocessing pipelines, and performance metrics such as precision, recall, and F1-score.
- Translation of model results into business insights by interpreting feature importance and identifying key churn drivers

House Pricing Prediction | Python, scikit-learn, pandas

- Development an end-to-end machine learning pipeline to predict house prices using the Ames Housing dataset using Gradient Boosting, among other models.
- Creation of a preprocessing pipeline, including custom missing-value handling with a custom transformer, feature encoding, scaling where appropriate, and justified metric selection (MAE, RMSE, and MAPE) based on the business problem.
- Evaluated model performance through quantitative metrics and residual analysis, identifying Gradient Boosting as the best-performing model and providing data-driven justification for model selection.

Portable Query Functions | Python, WebAssembly, SPARQL

- Creation of a pipeline enabling SPARQL queries to execute User-Defined Functions (UDFs) using isolated environments offered by WebAssembly
- Semantical definition of UDFs using the Function Ontology to make them more accessible and reusable following the FAIR principles

LANGUAGES

- Spanish – Native
- English – C1
- German – C1

CERTIFICATES

Kaggle Certificates

- Intermediate Machine Learning
- Data Cleaning
- Data Visualization
- Advanced SQL